

Industrial Internship Report on Prediction of Agriculture Crop Production in India

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Executive Summary

This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time.

My project was "**Prediction of Agriculture Crop Production in India**" using Machine Learning. The objective was to analyze historical agricultural production data (1993–2014) across various crops, units, and frequencies, and build a high-precision predictive regression model to estimate crop yield.

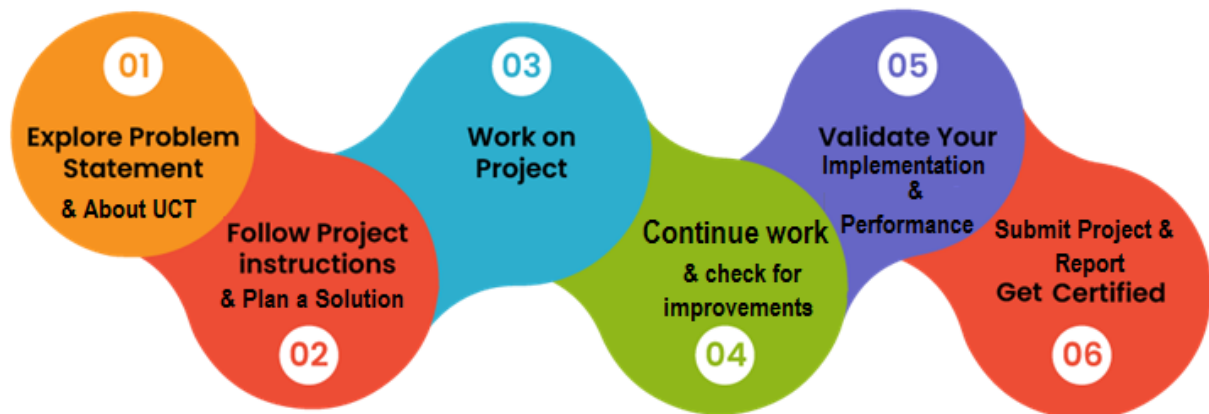
This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solutions for them. It was an overall great experience to have this internship.

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1 Preface

Over the 6-week internship, I worked on end-to-end data engineering, visual exploratory data analysis (EDA), and predictive modeling for agricultural dataset analysis. Relevant internships are vital for modern career development because they bridge the gap between academic machine learning concepts and real-world data constraints (such as handling unstructured nested files, wide time-series formats, and data leakage).



My project focused on transforming raw government agricultural data from a wide time-series matrix into a structured long-format dataset, training a RandomForestRegressor pipeline, and achieving accurate yield predictions across crop categories. I am grateful to upskill Campus and UniConverge Technologies Pvt Ltd for providing this platform, structured program execution, and continuous mentorship.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end etc.**



i. UCT IoT Platform (uct Insight)

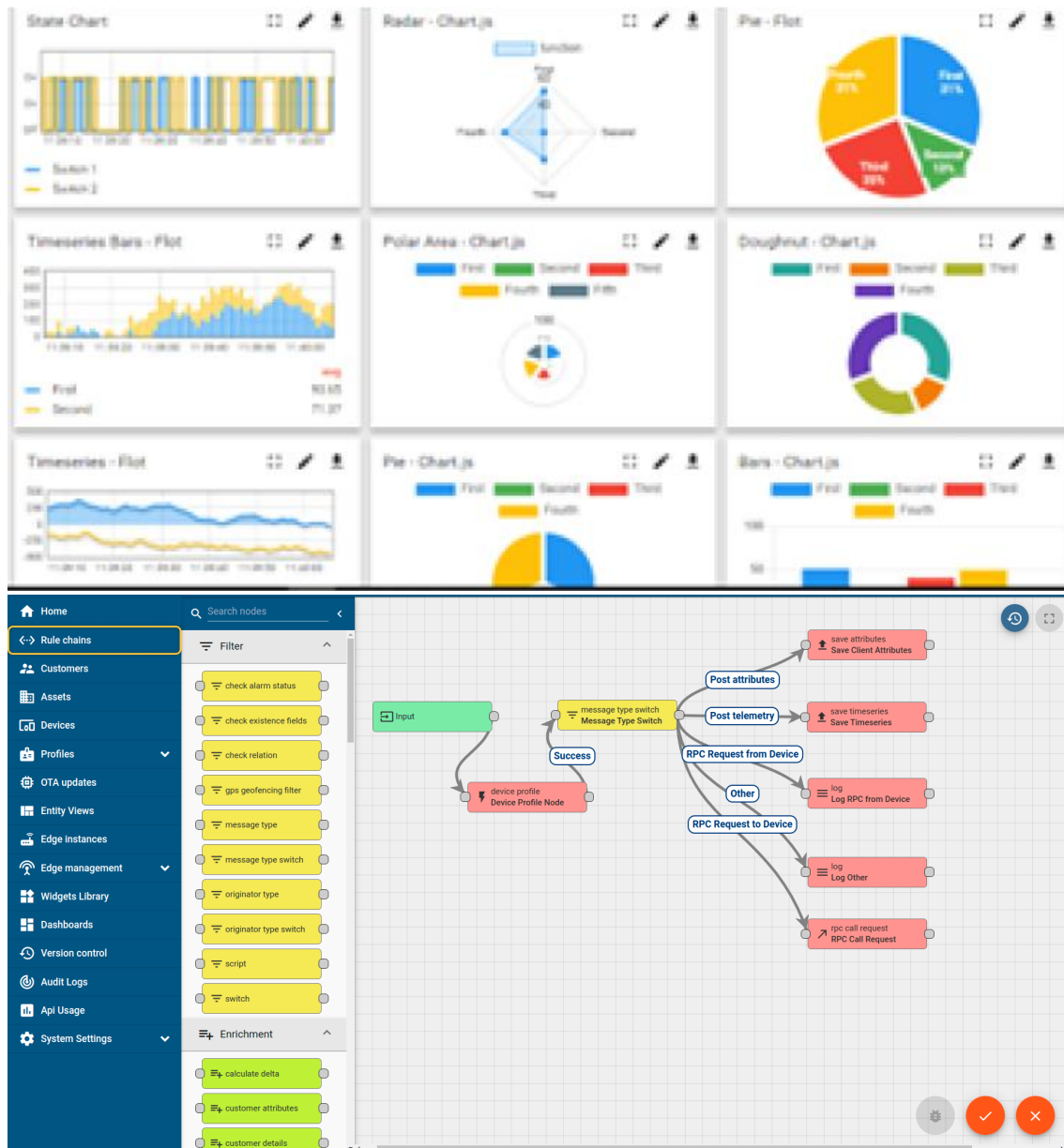
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification

- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



FACTORY
WATCH

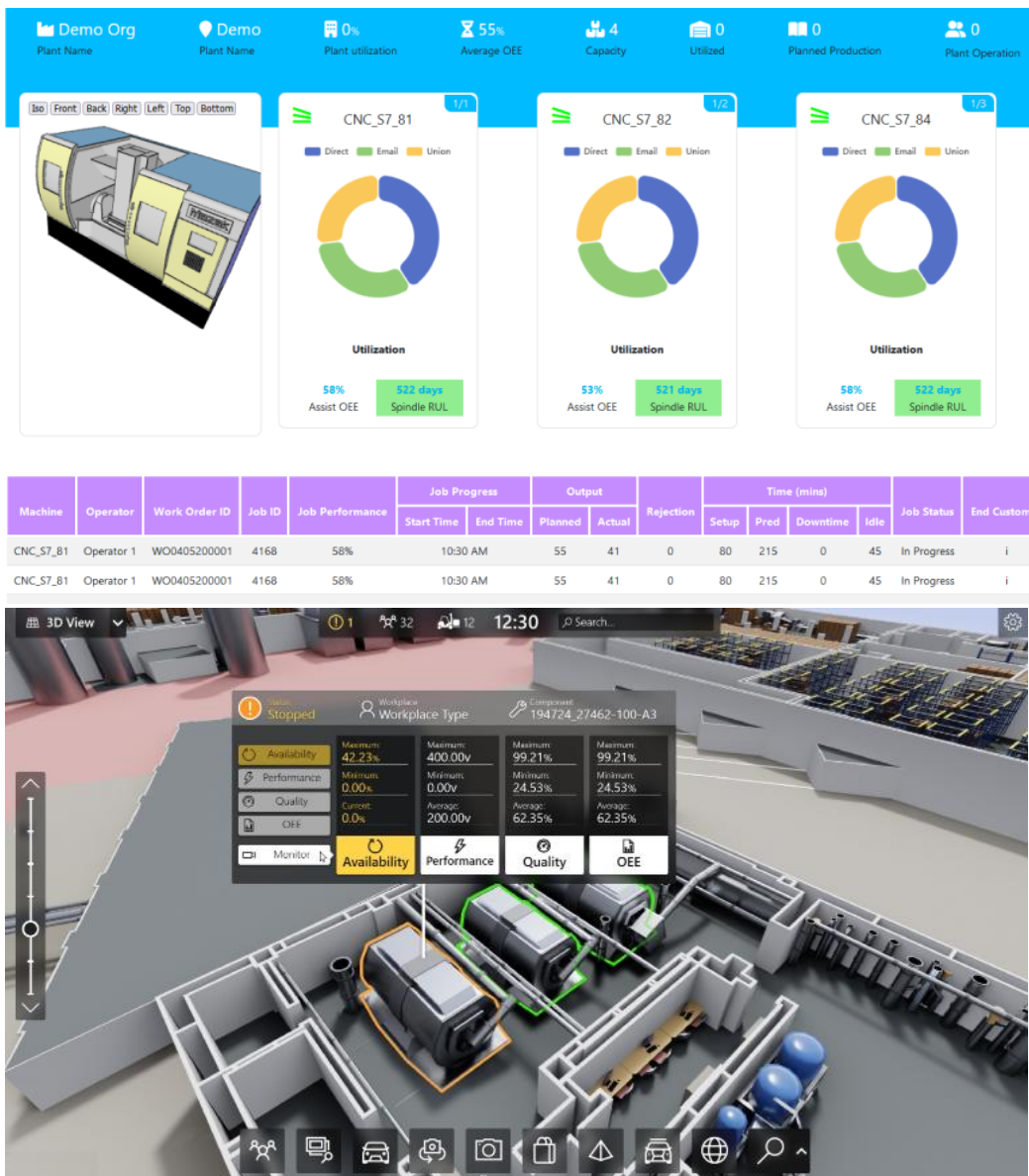
ii. Smart Factory Platform ()

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



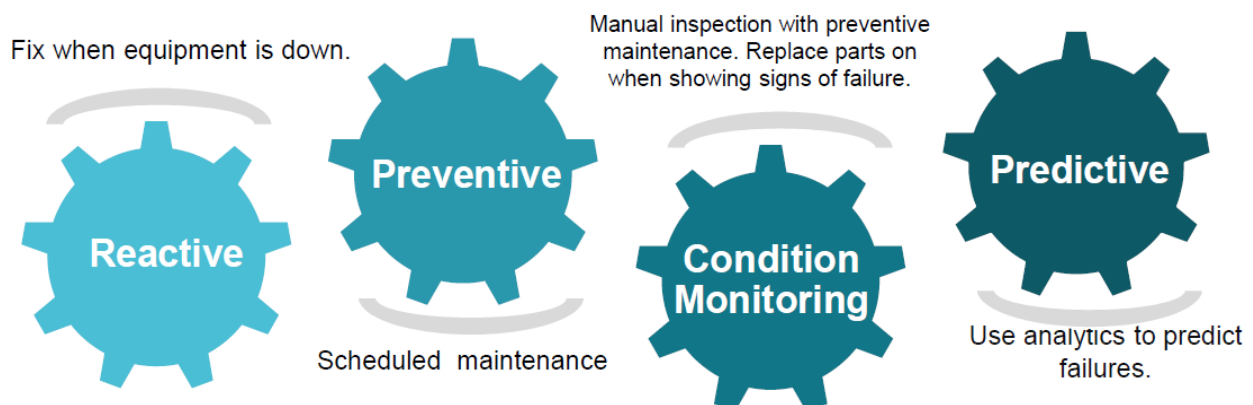


iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN technology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

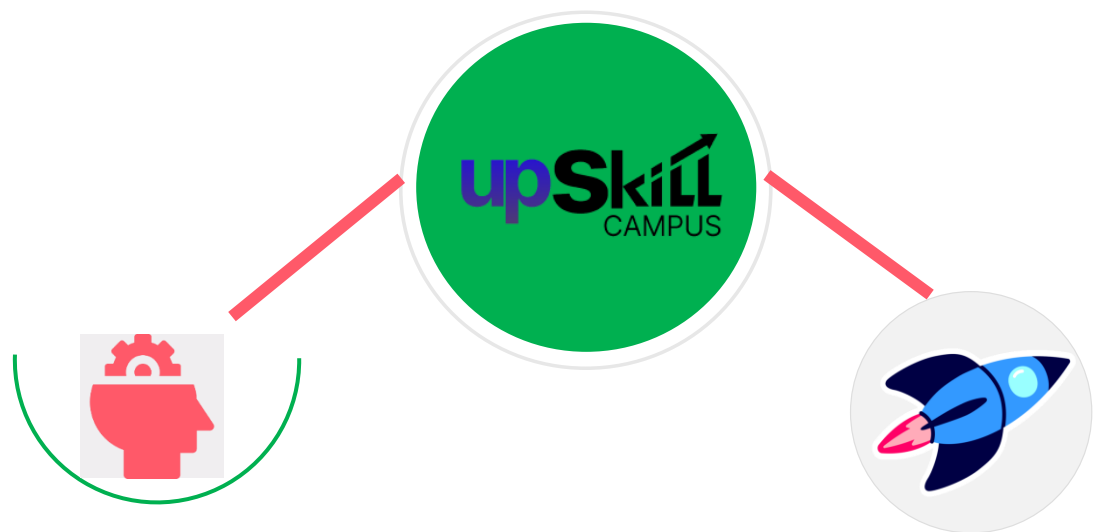
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

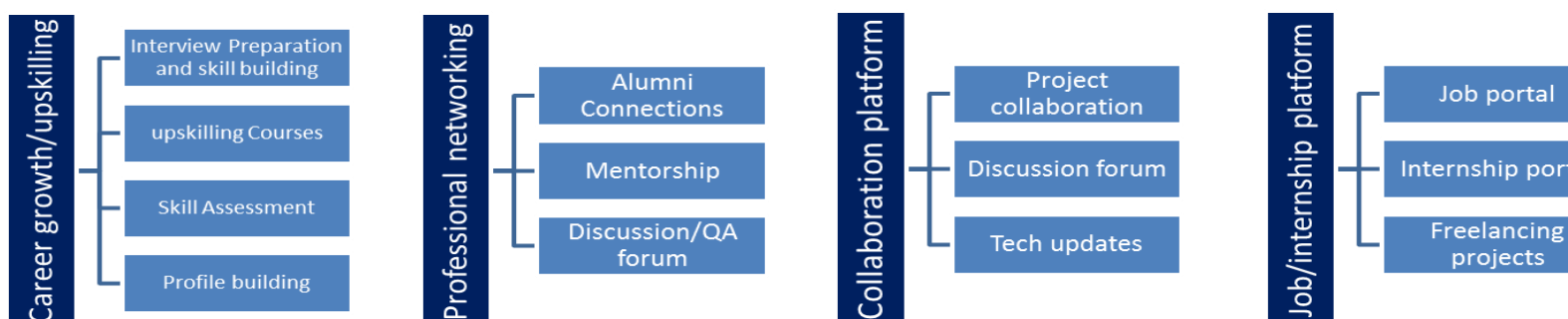
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

<https://www.upskillcampus.com/>



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- Gain practical experience working on real-world industrial datasets.
- Solve complex data challenges, including wide-format time-series unpivoting.
- Develop an end-to-end Machine Learning pipeline using scikit-learn and Python.
- Improve technical skills in data engineering, model evaluation, and documentation.

2.5 References

- Ministry of Agriculture & Farmers Welfare, Government of India (Historical Crop Yield Records 1993–2014).
- Scikit-Learn Documentation: RandomForestRegressor & ColumnTransformer Pipeline API.
- Seaborn & Matplotlib Visualization Libraries in Python.

2.6 Glossary

Term / Acronym	Definition
EDA	Exploratory Data Analysis PNG
MAE	Mean Absolute Error PNG
RMSE	Root Mean Squared Error PNG
R^2 Score	Coefficient of Determination (Model Accuracy Metric) PNG
OHE	One-Hot Encoding (Categorical Feature Scaling)

3 Problem Statement

In the assigned problem statement

Agricultural crop production estimation in India is complicated by multi-crop diversity, region-specific variations, and multi-year time-series structures. The primary objective of this project is to take historical crop production records (1993–2014) in wide format, reshape and clean the data into a normalized schema, and build a Machine Learning model capable of accurately predicting production volumes based on crop categories, measurement units, frequency, and time.

4 Existing and Proposed solution

Existing Solution & Limitations

Traditional agricultural reporting relies on static statistical tabulations or simple linear trend extrapolation across fixed years.

- **Limitations:** They fail to capture complex non-linear feature interactions between regional crop types, measurement units (Ton mn, Kg/ha), and multi-year temporal trends. Wide formats also increase the risk of temporal data leakage during model training.

Proposed Solution

We implement an automated Python data processing pipeline using pandas to unpivot wide yearly columns into a long-format schema (yielding **4,272 clean observation records**). A ColumnTransformer handles OneHotEncoder transformation for categorical metadata (Particulars, Unit, Frequency) and StandardScaler for Year. A RandomForestRegressor ensemble model is trained on this pipeline to deliver non-linear production predictions without data leakage.

4.1 Code submission (Github link)

<https://github.com/MukeshKumarGupta-234/upskillcampus/blob/main/CropProductionPrediction.ipynb>

4.2 Report submission (Github link) :

<https://github.com/MukeshKumarGupta-234/upskillcampus/blob/main/CropProductionPrediction.pdf>

5 Proposed Design/ Model

[Raw Dataset Archive (.zip)]

|



[Extraction & CSV Ingestion] —▶ [Wide-to-Long Melting (1993–2014)]

|



[Evaluation Plotting (Scatter/Bar)] ◀— [Random Forest Regressor Pipeline] ◀—
[ColumnTransformer Preprocessing]

6 Performance Test

6.1 Test Plan/ Test Cases

The reshaped dataset (4,272 records) was split into **80% Training Data** and **20% Test Data**. Model evaluation was conducted on unseen test data using three primary regression metrics:

- **Mean Absolute Error (MAE)**
- **Root Mean Squared Error (RMSE)**
- **R^2 Score (Coefficient of Determination)**

6.2 Test Procedure

- Execute nested extraction of raw zip archives.
- Clean column headers and melt matrix into long format.
- Perform Exploratory Data Analysis (Top 10 crop categories by production volume).
- Train RandomForestRegressor pipeline on training subset.
- Evaluate performance on testing subset and generate predicted vs. actual plot.
- Perform single-sample inference test for 2024 crop volume

6.3 Performance Outcome

- **EDA Visual:** Top crops identified Sugarcane and UP Wheat Volume as highest volume contributors.
- **Actual vs. Predicted Plot:** Scatter points showed high linear alignment along the $y = x$ diagonal across low, medium, and high yield scales.

7 My learnings

Through this industrial internship at upskill Campus & UCT, I learned:

- How to handle raw, unorganized time-series data and restructure wide matrices into long format suitable for ML pipelines.
- How to eliminate temporal data leakage in predictive models.
- Advanced usage of scikit-learn Pipeline and ColumnTransformer modules for reproducible feature scaling and encoding.
- Modern data visualization using Seaborn and version control management using GitHub.

8 Future work scope

- **Integration of Weather Data:** Incorporate regional rainfall, temperature, and soil moisture features to enhance real-world climate adaptation modeling.
- **Deep Learning Time-Series Models:** Experiment with LSTM or Temporal Fusion Transformers (TFT) for explicit sequential time-series forecasting.
- **Interactive Web App:** Deploy the trained pipeline model as a Streamlit or Flask web dashboard for real-time agricultural crop yield forecasting.